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Drug Discovery

Assignment: Molecular Docking

Submitted By: Siddiqa Rani
Submitted To: Dr Sadaf Zaidi

Assignment Report:

Molecular Docking of Favipiravir with SARS-CoV-2 NSP12 Polymerase (PDB ID: 6VWW)

1. Background:

Protein: SARS-CoV-2 NSP12 RNA-Dependent RNA Polymerase (RdRp) – PDB ID: 6VWW

The protein 6VWW corresponds to the **RNA-dependent RNA polymerase (RdRp)** of the SARS-Co V-2 virus. This enzyme is essential for viral replication and transcription, making it a prime target for antiviral drugs, especially against COVID-19. Targeting this protein can help in inhibiting the replicat ion cycle of the virus, thus potentially neutralizing its effects.

Why 6VWW?

- 6VWW is one of the **well-resolved crystal structures** of the viral polymerase, available in th e RCSB PDB.
- RdRp is **conserved** across coronaviruses, so inhibition may have broad-spectrum effects.
- It contains the **active site** that interacts with nucleotide analogs like Favipiravir.

Ligand: Favipiravir

Favipiravir is a **broad-spectrum antiviral drug**, initially developed for influenza. It inhibits viral RN A polymerase by mimicking nucleotides and incorporating into viral RNA, causing chain termination. Due to its known antiviral activity, it is studied for SARS-CoV-2 targeting.

2. Ligand and Protein Preparation

Step 1: Downloading the Protein (6VWW)

- Protein was downloaded from the RCSB Protein Data Bank in .pdb format.

Step 2: Removing Water Molecules and Non-Relevant Chains

- Opened 6VWW.pdb in **UCSF Chimera**.

- Also removed unnecessary ligands and ions using the structure control panel.
- Only the main polymerase chain was retained for docking.

Step 3: Adding polar Hydrogens and Charges

Step 4: Saving the Cleaned Protein

Step 5: Ligand Preparation (Favipiravir)

- Favipiravir was downloaded in .sdf format from PubChem. Therefore by using Chimera to wa s converted to PDB.
- The converted favipiravir.pdb file was used for further steps.

3. File Conversion to .pdbqt Format (AutoDock Requirement):

4. Google Colab: Docking Automation and Post-Processing

Google Colab was used to automate, execute, and analyze the molecular docking workflow without requiring local software installations. Here's a breakdown of the steps done inside Colab:

Step 1: Installing necessary libraries:

- Installs the **BioPython** package, which provides tools for reading and writing biological stru cture files (like PDB, FASTA).
- Also used later for extracting data from output files if needed.

Step 2: Uploading Required Files:

Step 2: Install Tools and Convert to .pdbqt Format

```
# Step 2.1: Install Vina, Open Babel, and Biopython
!apt-get install -y autodock-vina openbabel > /dev/null
!pip install biopython > /dev/null
```

```
# Step 2.2: Convert ligand to .pdbqt
!obabel favipiravin.pdb -O favipiravin.pdbqt --gen3d -p 7.4 -xh
```

```
1 molecule converted
```

```
# Step 2.3: Convert receptor to .pdbqt
!obabel PROTEIN.pdb -O PROTEIN.pdbqt -xr -xh
```

Step 3: Extracting Grid Center from PROTEINgrid.pdb

Step 3: Extract Grid Center from PROTEINgrid.pdb

```python
from Bio.PDB import PDBParser
import numpy as np
# Load the grid structure
parser = PDBParser(QUIET=True)
structure = parser.get_structure("GRID", "PROTEINgrid.pdb")
```

```python
# Extract coordinates
coords = [atom.coord for atom in structure.get_atoms()]
center = np.mean(coords, axis=0)
```

```python
# Store and display center values
center_x, center_y, center_z = center
print(f"Center X: {center_x:.2f}, Y: {center_y:.2f}, Z: {center_z:.2f}")
```

```
Center X: -71.77, Y: 21.87, Z: -23.98
```

To accurately dock a ligand, it is essential to define the 3D center of the binding pocket where dockin g will take place. In this step, the script uses **BioPython's PDBParser** to load a modified .pdb file (P ROTEINgrid.pdb) that represents the atoms of the active site. A list of atomic coordinates is extracted by looping over all atoms in the structure. Then, the script calculates the **average position (mean)** of all atoms across the x, y, and z axes using NumPy. This average represents the **geometric center** of th e binding site and is stored in the variables center_x, center_y, and center_z. These values will later be used to define the search space for AutoDock Vina.

Step 4: Creating the vina_config.txt Configuration File:

Step 4: Create vina_config.txt

```
with open("vina_config.txt", "w") as f:
    f.write(f"""\
receptor = PROTEIN.pdbqt
ligand = favipiravin.pdbqt
center_x = -71.77
center_y = 21.87
center_z = -23.98
size_x = 20
size_y = 20
size_z = 20
exhaustiveness = 8
""")
```

Once the center of the grid box is determined, a configuration file named vina_config.txt is created. T his file contains all the necessary parameters for running AutoDock Vina, formatted in a way that the software can read. The script writes the receptor and ligand file names (in .pdbqt format), the previous ly calculated center coordinates, and the size of the grid box in each dimension (x, y, and z). Addition ally, the parameter exhaustiveness is set to 8, which controls the thoroughness of the search — a highe r number of increases accuracy but also increases computation time. Automating this step ensures that the docking process remains consistent and reproducible.

Step 5: Running AutoDock Vina:

In this step, the actual molecular docking simulation is performed using AutoDock Vina. The Colab c

Step 5: Run Docking

```
!vina --config vina_config.txt --out docking_output.pdbqt
```

ell executes a command-line instruction that runs Vina with the previously created vina_config.txt file. This file provides all necessary input parameters including the search space and receptor/ligand files.

The result of the docking is saved in an output file called docking_output.pdbqt. This file contains the top-ranked binding poses of the ligand with associated scores (binding affinities), which can later be v isualized and analyzed. This step is the core of the docking workflow and provides predictions about h ow well the ligand fits and binds within the protein's active site.

```
Scoring function : vina
Rigid receptor: PROTEIN.pdbqt
Ligand: favipiravin.pdbqt
Grid center: X -71.77 Y 21.87 Z -23.98
Grid size  : X 20 Y 20 Z 20
Grid space : 0.375
Exhaustiveness: 8
CPU: 0
Verbosity: 1

Computing Vina grid ... done.
Performing docking (random seed: -995539190) ...
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
***************************************************
```

```
mode |   affinity | dist from best mode
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
-----+------------+----------+----------
   1       -4.365          0          0
   2       -4.158       2.45      3.748
   3        -4.13      1.821      2.847
   4        -4.03      2.472      3.153
   5       -3.995      2.022      2.176
   6       -3.754      19.15      20.28
   7       -3.684      19.35      20.15
   8       -3.511      19.16      20.04
   9       -3.473      3.508      4.205
```

This table shows:

- 9 different binding modes (poses) that Vina explored
- Their binding affinities (how strong the ligand binds to the protein)

- How different each pose is from the best pose (via RMSD)
- Pose 1 is the best binding conformation
- Affinity = -4.365 kcal/mol → This is moderate-to-weak binding (for drug-like molecules, we usually hope for < -6.0)
- Pose 2 and 3 are similar but not better

Conclusion

This assignment demonstrated the potential interaction between Favipiravir and the SARS-CoV-2 Rd Rp protein (6VWW), a key target for antiviral therapy. After preparing the protein and ligand, we calc ulated the binding site center and created a docking configuration using BioPython and Google Colab. Docking was carried out using AutoDock Vina, which predicted a moderate binding affinity, indicati ng Favipiravir's possible inhibitory effect on the viral polymerase. Overall, this highlights the value of computational tools in supporting early-stage drug discovery and screening.